

**AN INTELLIGENT RFID AND CLOUD-BASED INVENTORY
TRACKING SYSTEM FOR HEALTHCARE SUPPLY CHAINS**

A PROJECT REPORT

Submitted by

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ABSTRACT

In most hospital pharmacies and medical stores, patient care activities are critical but are done in environments where traditional methods for maintaining patient care item inventories are either manual or semi-automated. This has resulted in stock-outs, expired medications, and losses. In past research, it has been proven that using RFID technology allows for real-time identification and monitoring of medications in real time, thus making it possible to properly identify stock levels. More advanced are solutions based on Internet of Things technology in pharmacies. Such solutions go beyond basic RFID technology by allowing for continuous data transmission and monitoring. Despite all current advancements in this field, most of these solutions are response-based and therefore do not have any predictive analytical intelligence capabilities for demand forecasting and automatic management of patient care item inventory levels. Scalability and decision-making are two other significant concerns now in using this technology in multiple-hospital pharmacies. To overcome such shortcomings in current solutions, in this article, we introduce an Internet of Things-based Cloud AI-integrated real-time RFID system for patient care item inventory management in hospital pharmacies and medical store environments.

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Inventory management in hospital pharmacies plays a critical role in ensuring the availability of medicines and medical supplies required for patient care. Traditional inventory management systems in hospitals often rely on manual records or barcode-based tracking methods. These systems are prone to human errors, delays in updating stock information, and difficulties in tracking medicine movement across departments.

With the advancement of technologies such as **Radio Frequency Identification (RFID)**, **Internet of Things (IoT)**, and **Cloud Computing**, healthcare inventory management can be automated and monitored in real time. RFID technology allows automatic identification of medicines through RFID tags attached to each item, while RFID readers capture the information without requiring manual scanning.

In this project, an **RFID and cloud-based inventory tracking system** is developed to monitor the movement and availability of medicines in hospital pharmacies. RFID readers detect tagged medicines and send the data through an IoT-enabled microcontroller to a cloud platform. The cloud system stores the data and provides real-time access to inventory information through a web dashboard.

This system helps hospital administrators monitor stock levels, track medicine usage, reduce manual errors, and prevent shortages or wastage due to expired

medicines. By integrating RFID technology with cloud-based monitoring, the proposed system provides a **reliable, automated, and efficient solution for healthcare inventory management.**

1.2 SCOPE OF THE PROJECT

The scope of this project includes the development of an automated inventory monitoring system for hospital pharmacies using RFID and cloud technology. The system focuses on improving efficiency and accuracy in tracking medicines and managing stock levels.

The major scope of the project includes:

- Automated identification of medicines using RFID tags and RFID readers.
- Real-time monitoring of inventory movement in hospital pharmacies.
- Cloud-based storage of inventory data for centralised access.
- Automatic updating of stock information whenever medicines are added or removed.
- Remote monitoring of inventory status through a web-based dashboard.
- Reduction of manual errors in medicine tracking.
- Prevention of medicine shortages and better inventory planning.

The system can be implemented in hospitals, medical stores, and pharmaceutical warehouses to improve inventory transparency and operational efficiency.

1.3 OBJECTIVES

The main objective of this project is to develop an **RFID and cloud-based intelligent inventory tracking system** for managing medicines in hospital pharmacies.

The specific objectives include:

- To automate medicine identification using RFID technology.
- To monitor medicine movement in real time within the hospital supply chain.
- To transmit inventory data to a cloud server using IoT connectivity.
- To provide a centralized system for monitoring stock levels.
- To reduce manual record maintenance and associated human errors.
- To improve efficiency in inventory auditing and stock management.
- To ensure availability of medicines and reduce losses due to mismanagement or expiry.

CHAPTER 2

LITERATURE SURVEY

The literature survey provides an overview of existing research and technologies related to RFID-based inventory management, IoT-enabled healthcare monitoring, and cloud-based healthcare systems. Several research studies have focused on improving healthcare operations using modern technologies such as RFID, cloud computing, and IoT systems. The following papers highlight significant contributions in these areas.

2.1 PAPER 1

Title: Blockchain and Fog Computing in IoT-Driven Healthcare Services for Smart Cities

This research paper discusses the integration of blockchain technology and fog computing in IoT-based healthcare services. The primary objective of the study is to enhance the security, efficiency, and reliability of healthcare systems operating in smart cities. The proposed framework uses IoT devices to collect healthcare data from various medical sensors and monitoring devices. These devices continuously generate large volumes of patient-related data, which must be processed and stored securely.

Fog computing is introduced as an intermediate processing layer between IoT devices and cloud servers. Instead of sending all the data directly to the cloud, the fog layer performs preliminary data processing and filtering. This reduces network latency and improves the speed of healthcare services. In addition, blockchain technology is used to ensure secure data sharing and protect sensitive patient information from unauthorized access.

The study highlights that the combination of blockchain and fog computing improves data security and system reliability in healthcare environments.

However, implementing blockchain-based systems can be computationally intensive and may require additional infrastructure.

The findings of this research emphasize the importance of secure and efficient data management in healthcare systems. Although the proposed approach mainly focuses on patient health monitoring, the concepts of secure data transmission and distributed processing can also be applied to healthcare inventory management systems.

2.2 PAPER 2

Title: IoT-Based Warehouse Management System Leveraging RFID and Cloud Technologies

This research paper focuses on the implementation of an IoT-based warehouse management system that uses RFID technology and cloud platforms to monitor inventory in real time. Traditional warehouse systems rely heavily on manual processes such as barcode scanning and manual stock verification. These approaches are often inefficient and prone to human errors.

The proposed system utilizes RFID tags attached to inventory items, which can be automatically detected using RFID readers. Unlike barcode systems, RFID technology does not require direct line-of-sight scanning and can detect multiple items simultaneously. This significantly improves the speed and efficiency of inventory tracking.

The RFID readers are connected to IoT-enabled devices that transmit collected data to a cloud server. The cloud platform stores the inventory information and provides real-time updates through a web interface. This enables administrators to monitor warehouse operations remotely and manage stock levels effectively.

The research demonstrates that the integration of RFID technology with IoT communication and cloud storage greatly improves inventory visibility and

reduces operational costs. Automated tracking systems minimize manual intervention and improve the accuracy of stock records.

The ideas presented in this paper are directly applicable to healthcare supply chains, where accurate tracking of medicines and medical supplies is essential. Implementing RFID-based systems in hospital pharmacies can help prevent stock shortages, reduce medicine wastage, and improve inventory control.

2.3 PAPER 3

Title: Scalable AI Solutions for IoT-Based Healthcare Systems using Cloud Platforms

This research explores the use of cloud computing and artificial intelligence technologies in IoT-based healthcare systems. The study focuses on how cloud infrastructure can support large-scale healthcare applications that generate massive amounts of data from connected medical devices.

IoT devices such as wearable health monitors, smart medical equipment, and hospital sensors continuously collect patient-related information. Cloud platforms provide the necessary computing power and storage capacity to manage and analyze this data efficiently. By storing healthcare data in centralized cloud databases, medical professionals can access patient information remotely and monitor health conditions in real time.

The research also highlights the potential of artificial intelligence algorithms to analyze healthcare data and identify patterns related to patient conditions. AI models can assist in disease prediction, treatment planning, and healthcare decision-making.

Although the primary focus of the study is on patient health monitoring, the underlying cloud infrastructure can also support other healthcare applications such as hospital resource management and inventory monitoring. Cloud-based

systems provide scalability, allowing healthcare institutions to manage data from multiple departments and locations efficiently.

The concepts discussed in this research emphasize the importance of cloud-based systems for handling large datasets and enabling remote monitoring capabilities. These principles are highly relevant for implementing modern healthcare inventory management systems that rely on IoT devices and cloud connectivity.

2.4 PAPER 4

Title: Cloud-Based Healthcare Monitoring Systems

This paper discusses the role of cloud computing in modern healthcare systems. Cloud platforms provide a centralized infrastructure for storing and managing large volumes of healthcare data generated by IoT devices and medical equipment. By using cloud services, healthcare providers can access patient information remotely, enabling better monitoring and faster decision-making.

The system architecture typically involves data collection through sensors, transmission via IoT networks, and storage in cloud databases. The cloud platform processes the data and provides meaningful insights through dashboards and visualization tools. This allows healthcare professionals to monitor patient conditions in real time and respond quickly to emergencies.

The study highlights the scalability and flexibility of cloud computing, which makes it suitable for large-scale healthcare applications. Although the paper focuses on patient monitoring, the same cloud-based approach can be applied to inventory management systems. By storing inventory data in the cloud, hospitals can achieve centralized control, remote access, and efficient data management.

2.5 PAPER 5

Title: RFID-Based Smart Inventory System for Hospitals

This research paper explores the use of RFID technology in hospital inventory management systems. Hospitals handle a large number of medical supplies,

equipment, and medicines, making manual tracking difficult and error-prone. RFID technology provides an automated solution by enabling real-time tracking of items through unique identification tags.

In the proposed system, RFID tags are attached to medical items, and RFID readers are installed at various locations within the hospital. Whenever an item is moved, the RFID reader captures the tag information and updates the inventory database automatically. This reduces the need for manual record-keeping and improves accuracy in tracking inventory.

The system significantly reduces time required for inventory audits and ensures better availability of medical supplies. However, the implementation cost of RFID infrastructure can be a limitation. Despite this, the benefits of automation, accuracy, and efficiency make RFID-based systems highly effective for healthcare inventory management.

2.6 PAPER 6

Title: IoT-Enabled Smart Healthcare System

This paper presents an IoT-based smart healthcare system designed to monitor patient health using connected devices and sensors. The system collects real-time data such as temperature, heart rate, and other vital signs, which are transmitted to a central server for analysis and monitoring.

The IoT architecture enables seamless communication between devices, allowing healthcare professionals to access patient data remotely. This improves the efficiency of healthcare services and enables timely medical interventions. The system also supports continuous monitoring, which is particularly useful for patients with chronic conditions.

Although the paper focuses on patient monitoring, the concept of real-time data collection and remote access can be extended to inventory systems. By applying IoT

technology to inventory management, hospitals can monitor stock levels continuously and ensure the availability of essential medicines.

2.7 PAPER 7

Title: RFID Technology in Supply Chain Management

This study analyzes the impact of RFID technology on supply chain management. RFID systems enable automatic identification and tracking of goods throughout the supply chain, improving visibility and efficiency. Unlike traditional methods, RFID does not require manual scanning, which reduces human errors and speeds up operations.

The system involves attaching RFID tags to products and using RFID readers to capture data at various stages of the supply chain. The collected data is stored in a central database, allowing organizations to monitor the movement of goods in real time. This improves coordination between different stages of the supply chain and enhances overall efficiency.

The research highlights that RFID technology improves inventory accuracy, reduces losses, and enhances decision-making. These benefits are particularly important in healthcare supply chains, where accurate tracking of medicines is critical. The study supports the use of RFID technology in developing efficient inventory management systems for hospitals.

2.8 PAPER 8

Title: Smart Pharmacy Management System using IoT

This paper presents a smart pharmacy management system that utilizes Internet of Things (IoT) technology to automate the monitoring and management of medicines in pharmacies. Traditional pharmacy systems rely heavily on manual record-keeping, which can lead to inefficiencies and errors. The proposed system uses sensors and connected devices to track medicine stock levels and update records automatically.

In this system, IoT devices are used to continuously monitor the availability of medicines and transmit data to a centralized server. The server processes the data and provides real-time updates through a user interface. The system can also generate alerts when stock levels fall below a certain threshold, helping pharmacy staff take timely action to restock medicines.

The study highlights that IoT-based pharmacy systems improve operational efficiency, reduce manual workload, and enhance accuracy in inventory tracking. These features are highly relevant to hospital pharmacy systems, where maintaining accurate and up-to-date inventory information is essential for patient care.

2.9 PAPER 9

Title: Cloud-Based Inventory Management Systems

This research focuses on the use of cloud computing technology for managing inventory systems. Traditional inventory systems often store data locally, which limits accessibility and scalability. Cloud-based systems, on the other hand, provide centralized storage and allow users to access data from any location with internet connectivity.

The system architecture involves collecting inventory data from various sources and storing it in a cloud database. The cloud platform processes the data and provides real-time updates through dashboards and reporting tools. This enables administrators to monitor inventory levels, track item movement, and make informed decisions.

The study emphasizes that cloud-based systems improve data accessibility, scalability, and system reliability. In healthcare environments, cloud-based inventory systems can help manage multiple pharmacy locations efficiently and ensure that critical medicines are always available when needed.

2.10 PAPER 10

Title: Automated Inventory Tracking using RFID and IoT

This paper proposes an automated inventory tracking system that combines RFID

technology with IoT communication. The system aims to eliminate manual intervention in inventory management by automating the process of data collection and transmission.

RFID tags are attached to inventory items, and RFID readers are used to detect these tags automatically. The collected data is processed by an IoT-enabled microcontroller and transmitted to a cloud server. The cloud platform stores the data and provides real-time updates to users through a web interface.

The research demonstrates that integrating RFID and IoT technologies significantly improves inventory accuracy and efficiency. The system reduces human errors, speeds up data processing, and enables real-time monitoring. This approach is highly suitable for healthcare inventory systems, where accuracy and reliability are critical.

CHAPTER 3

METHODOLOGY

3.1 EXISTING SYSTEM

In many hospital pharmacies, inventory management is performed using manual record books or barcode systems. Pharmacists manually update stock information whenever medicines are received or dispensed. Periodic stock verification is also carried out to identify shortages or expired medicines.

Although barcode systems provide a certain level of automation, they still require manual scanning and human intervention. These traditional methods become inefficient when managing large quantities of medicines and medical supplies. As a result, maintaining accurate inventory records becomes difficult and time-consuming.

3.1.1 DISADVANTAGES

- Manual record keeping is time-consuming.
- Higher chances of human errors in data entry.
- Lack of real-time inventory visibility.
- Difficulty in tracking medicine movement.
- Increased risk of medicine expiry and stock shortages.

3.2 PROPOSED SYSTEM

The proposed system introduces an RFID-based automated inventory tracking solution for hospital pharmacies. In this system, each medicine package is attached with an RFID tag containing a unique identification number.

RFID readers installed in storage shelves and dispensing areas automatically scan the tags whenever medicines are moved.

The RFID reader captures the tag information and sends it to the ESP32 microcontroller. The microcontroller processes the data and transmits it to a cloud server using Wi-Fi connectivity. The cloud platform stores the inventory information and updates the database in real time.

Hospital administrators can access this information through a web-based dashboard. This allows them to monitor stock levels, detect shortages, and track medicine usage efficiently.

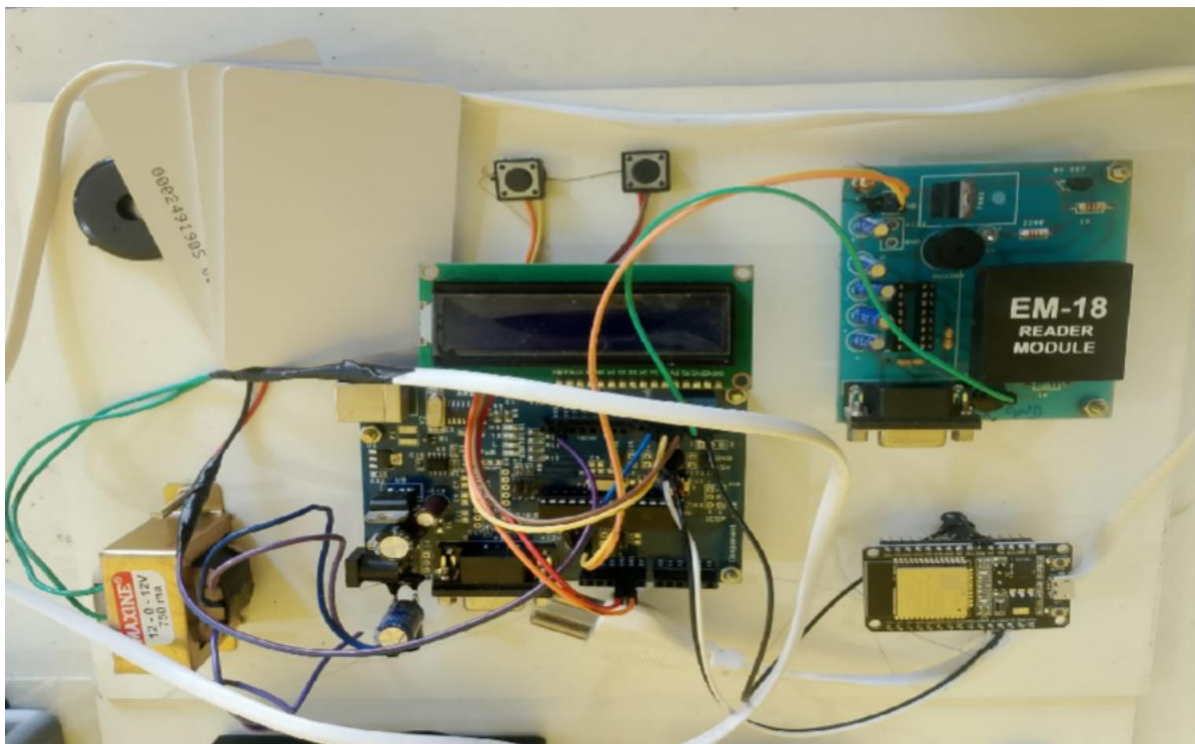


Fig 3.2

3.2.1 ADVANTAGES

- Automated inventory tracking.
- Real-time stock monitoring.
- Reduced manual errors.
- Faster inventory auditing.
- Improved supply chain transparency.
- Remote monitoring through cloud platforms.

3.3 SYSTEM CONFIGURATION

3.3.1 System Integration

The system integrates RFID hardware components with IoT communication and cloud-based storage. RFID readers capture medicine information, which is processed by the ESP32 microcontroller and transmitted to the cloud server.

3.3.2 Data Acquisition

RFID readers automatically capture identification information stored in RFID tags attached to medicines.

3.3.3 Data Processing and Transmission

The ESP32 microcontroller processes the captured RFID data and sends it to the cloud server using Wi-Fi connectivity.

3.3.4 Cloud Data Storage

The cloud server stores inventory data in a centralized database, ensuring secure storage and easy access to inventory information.

3.3.5 User Interface

A web dashboard provides real-time visualization of inventory status. Hospital administrators can monitor stock levels and receive alerts regarding inventory updates.

3.4 ALGORITHM

Algorithm: RFID-Based Inventory Tracking System

Step 1: Start the system.

Step 2: Initialize ESP32 microcontroller, RFID reader, LCD display, and Wi-Fi module.

Step 3: Wait for an RFID tag to be detected by the RFID reader.

Step 4: Read the RFID tag ID.

Step 5: Send the tag ID to the ESP32 microcontroller. Step

6: Display tag information on the LCD display.

Step 7: Transmit the tag data to the cloud server through Wi-Fi.

Step 8: Update the inventory database on the server.

Step 9: Display updated inventory status on the web dashboard.

Step 10: Repeat the process for the next RFID scan.

Features of RFID Sensor

- Operating Voltage: 5V.
- Contactless identification technology.
- Unique identification for each tag.
- Fast scanning capability.
- Suitable for real-time tracking applications.

RFID technology eliminates the need for manual scanning and improves the accuracy of inventory tracking.

CHAPTER 4

SYSTEM DESIGN

4.1 ARCHITECTURE

The architecture of the proposed system is designed to provide an efficient and automated solution for medicine inventory management in hospital pharmacies and medical stores. The system integrates RFID technology, IoT devices, cloud computing, and a web-based dashboard to monitor and manage medicine stock in real time.

In this system, RFID tags are attached to each medicine package. These RFID tags contain a unique identification number which helps in identifying and tracking individual medicines. Whenever a medicine is scanned or moved, the RFID reader reads the tag information automatically.

The RFID reader is connected to the ESP32 microcontroller which acts as the central processing unit of the system. The microcontroller receives the tag data from the RFID reader and processes it. Using the built-in Wi-Fi capability of the ESP32, the processed data is transmitted to the cloud server.

The cloud server stores and manages the inventory information in a centralized database. It also performs data processing and analytics. The stored information can then be accessed through a web dashboard that allows administrators or pharmacy staff to monitor stock levels, check expiry dates, and track medicine movement.

Additionally, the system provides alerts for low stock levels and near expiry medicines, helping hospital staff take timely action. The overall architecture ensures real-time monitoring, reduced manual work, improved accuracy, and better inventory management in hospital pharmacies.

4.2 BLOCK DIAGRAM

The block diagram of the proposed system illustrates the flow of data between the hardware components and the cloud platform.

The system begins with the power supply which provides the necessary electrical power to all the hardware components. The ESP32 microcontroller acts as the main controller that manages the communication between the devices.

The RFID reader scans the RFID tags attached to the medicine packages and sends the tag data to the microcontroller. The microcontroller processes this information and transmits it to the cloud server through Wi-Fi connectivity.

The cloud server stores the received data and processes it for inventory management. The processed information is then displayed through a web dashboard where users can monitor the medicine stock and receive alerts.

An LCD display is also connected to the system to show real-time status such as tag detection and data transmission confirmation.

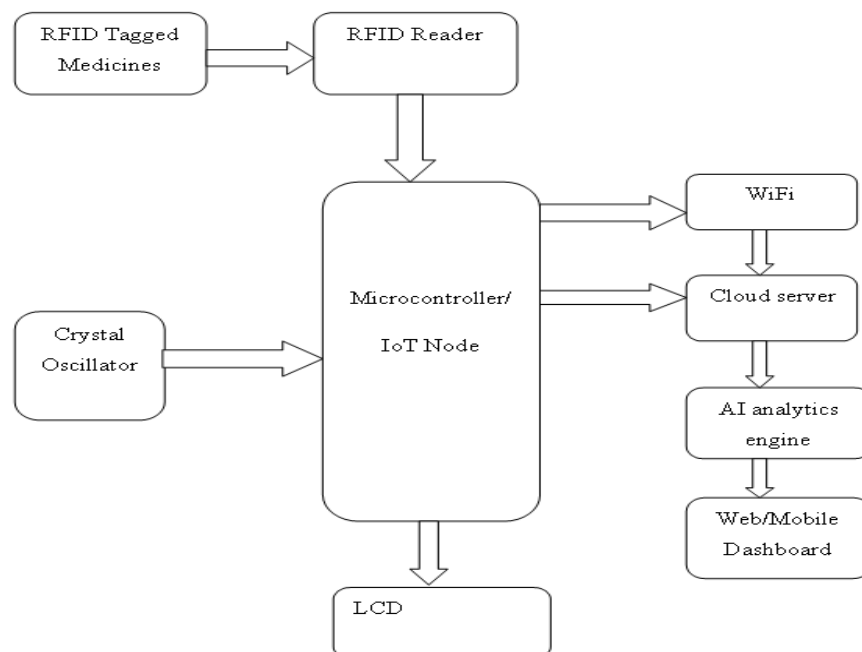


Fig4.2 Block Diagram of the Proposed System

4.3 TOOLS USED

The implementation of the proposed system requires both hardware components and software tools. These tools help in building the hardware setup, programming the microcontroller, and managing the cloud-based inventory system.

4.3.1 Hardware

The following hardware components are used in the implementation of the RFID-based inventory tracking system.

1. 2ESP32 Microcontroller

The ESP32 microcontroller acts as the core component of the system. It receives data from the RFID reader, processes the information, and sends it to the cloud server using Wi-Fi connectivity. It controls the communication between all hardware components and ensures smooth operation of the system.



Fig4.3.1

2. RFID Reader

The RFID reader is used to scan RFID tags attached to medicine packages. It reads the unique identification number stored in each RFID tag and sends the data to the microcontroller for further processing.



Fig4.3.2

3. RFID Tags

RFID tags are attached to each medicine package. These tags store unique identification information which helps in automatically tracking medicines without manual data entry.

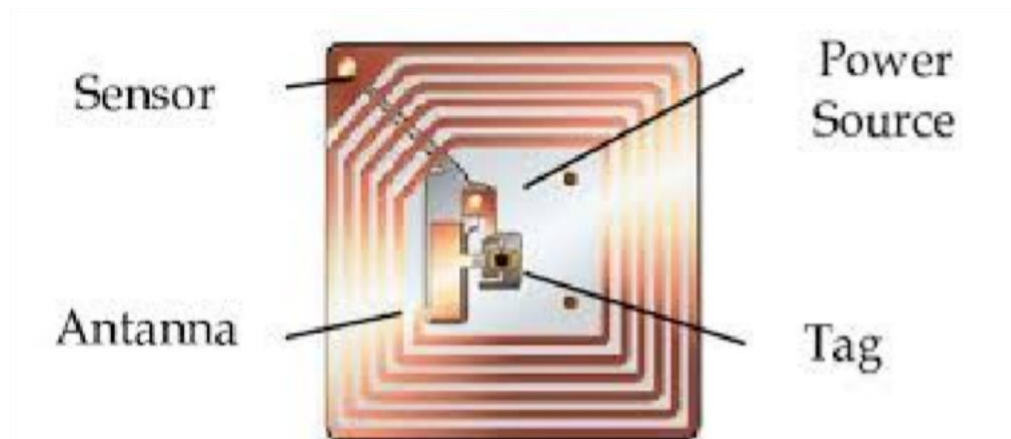


Fig4.3.3

4. LCD Display

The LCD display is used to show system status and information such as successful tag detection and confirmation of data transmission. It helps users monitor the functioning of the system locally.



Fig4.3.4

5. Power Supply

The power supply provides the required electrical power to all components of the system including the microcontroller, RFID reader, and display module.



Fig4.3.5

4.3.2 Software

The following software tools are used for developing and managing the system.

1. Embedded C

Embedded C programming language is used to write the code for the ESP32 microcontroller. It controls the interaction between the RFID reader, Wi-Fi module, and other hardware components.

2. Arduino IDE

Arduino IDE is used for writing, compiling, and uploading the program to the ESP32 microcontroller. It provides the development environment required for coding and testing the hardware components.

3. Cloud Server (PHP / Web Dashboard)

A cloud server is used to store and manage the inventory data received from the microcontroller. PHP is used for backend processing, while a web dashboard is developed to display inventory information such as medicine stock levels, expiry alerts, and real-time updates.

CHAPTER 5

HARDWARE DESCRIPTION

5.1 MICROCONTROLLER (ESP32)

The ESP32 microcontroller is the main processing unit of the proposed RFID-based inventory tracking system. It controls the overall operation of the system and coordinates communication between different hardware components such as the RFID reader, LCD display, and Wi-Fi communication module.

The ESP32 receives data from the RFID reader whenever a medicine with an RFID tag is scanned. After receiving the tag information, the ESP32 processes the data and sends it to the cloud server through Wi-Fi connectivity. The ESP32 also updates the LCD display to show the current status of the system.

The ESP32 microcontroller is widely used in IoT applications because it provides built-in Wi-Fi and Bluetooth connectivity, enabling easy integration with cloud platforms and remote monitoring systems.

5.1.1 Specifications of ESP32

- Dual-core 32-bit processor.
- Operating frequency up to 240 MHz.
- Built-in Wi-Fi and Bluetooth connectivity.
- 520 KB SRAM memory.
- Multiple GPIO pins for hardware interfacing.
- Supports SPI, I2C, and UART communication protocols.
- Low power consumption.

These features make the ESP32 suitable for IoT-based real-time monitoring systems.

A stable power supply ensures proper functioning of the ESP32 microcontroller, RFID reader, and LCD display.

5.2 POWER SUPPLY

The power supply provides the required electrical energy to operate all system components. A stable power supply ensures proper functioning of the ESP32 microcontroller, RFID reader, and LCD display.

In this system, a regulated 12V adapter is used to supply power to the hardware components. The adapter converts AC voltage from the mains supply into DC voltage suitable for electronic circuits.

5.2.1 Features

- Input Voltage: 220V – 230V AC.
- Output Voltage: 12V DC.
- Output Current: 1A.
- Reliable and stable power supply.
- Low power consumption.

The regulated power supply prevents voltage fluctuations that could damage the hardware components.

5.3 LCD DISPLAY (16×2)

information about the RFID scanning process. It consists The 16×2 LCD display is used to display system messages and real-time of two rows and sixteen columns, allowing it to display up to 32 characters at a time.

The LCD display is connected to the ESP32 microcontroller through digital input/output pins. Whenever an RFID tag is scanned, the microcontroller updates the LCD display to show relevant information such as successful tag detection or system status.

5.3.1 Features

- Operating Voltage: 5V.
- Display Format: 16 characters × 2 lines.
- Low power consumption.
- Easy interface with microcontrollers.
- Clear and simple user interface.

The LCD helps users quickly understand the system status without accessing the cloud dashboard.

5.4 INTERNET OF THINGS (IoT)

The Internet of Things (IoT) enables communication between physical devices and cloud-based systems through the internet. In this project, IoT technology allows the RFID inventory system to transmit data from the ESP32 microcontroller to a cloud server.

Whenever a medicine is scanned using the RFID reader, the ESP32 sends the information to the cloud platform using Wi-Fi connectivity. This allows hospital administrators to monitor inventory information remotely through a web-based dashboard.

IoT technology improves inventory visibility and enables real-time monitoring of medicine stock levels.

5.5 WEB SERVER

The web server is responsible for receiving and storing inventory data transmitted from the ESP32 microcontroller. It maintains a database that stores information such as medicine ID, scanning time, and inventory status.

The web server processes incoming data and updates the inventory database accordingly. A web dashboard connected to the server allows administrators to view the current inventory status through an internet browser.

5.5.1 Controlling Section

The controlling section of the web server performs the following tasks:

- Receives RFID data from the ESP32 microcontroller.
- Stores inventory information in the database.
- Updates the web dashboard in real time.
- Allows remote monitoring of stock levels.
- Provides centralized inventory management.

5.6 RFID READER AND RFID TAGS

system RFID technology is the core component of the proposed inventory tracking.

RFID Tags

RFID tags are attached to medicine packages. Each tag contains a unique identification number that represents a specific medicine in the inventory database.

RFID Reader

The RFID reader detects RFID tags when they come within its reading range. Once the tag is detected, the reader sends the tag ID to the ESP32 microcontroller for processing.

CHAPTER 6

SOFTWARE COMPONENTS

The software component of the proposed system plays a crucial role in controlling the hardware devices and managing the inventory data. The system software is responsible for programming the microcontroller, processing RFID data, and sending the information to the cloud platform for storage and monitoring.

In this project, the software tools used include Arduino IDE for programming the ESP32 microcontroller, Embedded C for writing the program logic, and a cloud database with a web dashboard for storing and displaying the inventory data.

6.1 ARDUINO IDE

The Arduino Integrated Development Environment (IDE) is used to develop and upload the program for the ESP32 microcontroller. It provides a simple and user-friendly interface for writing, editing, compiling, and uploading code to embedded hardware devices.

Arduino IDE supports programming languages such as Embedded C and C++. It also provides a wide range of built-in libraries that make it easier to interface with hardware components like RFID readers, LCD displays, Wi-Fi modules, and sensors.

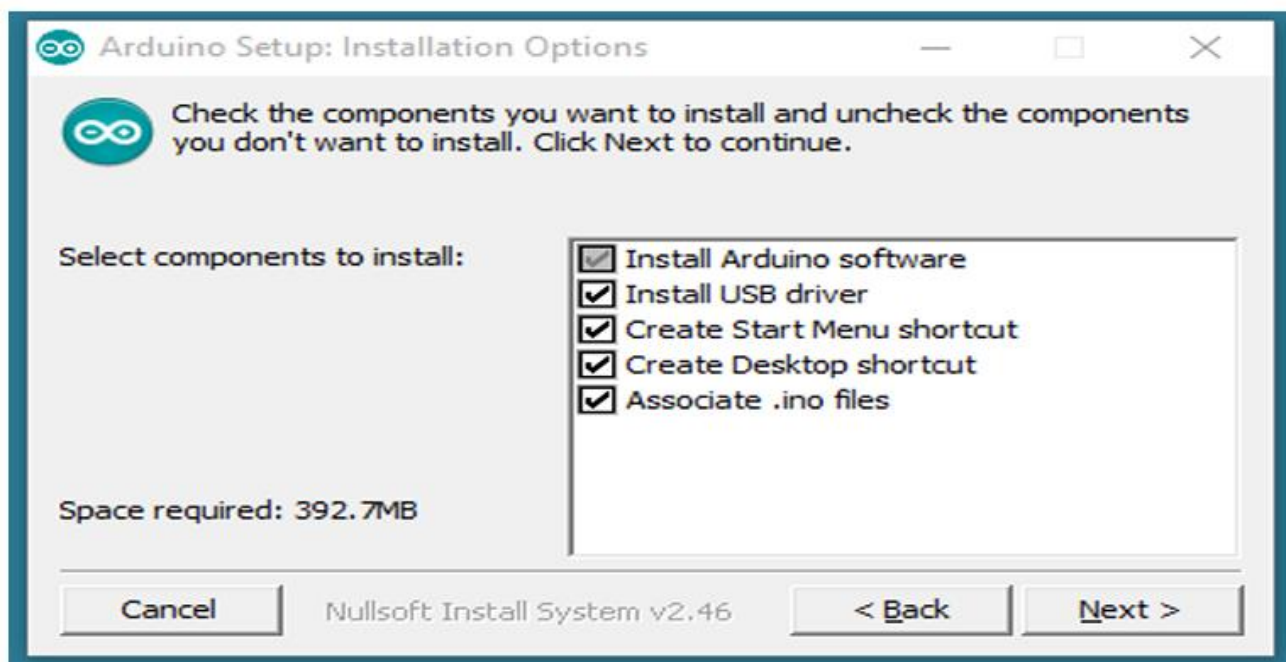
In this project, Arduino IDE is used to write the program that controls the RFID reader, processes the scanned tag information, and sends the data to the cloud server through Wi-Fi connectivity.

The steps involved in programming using Arduino IDE are as follows:

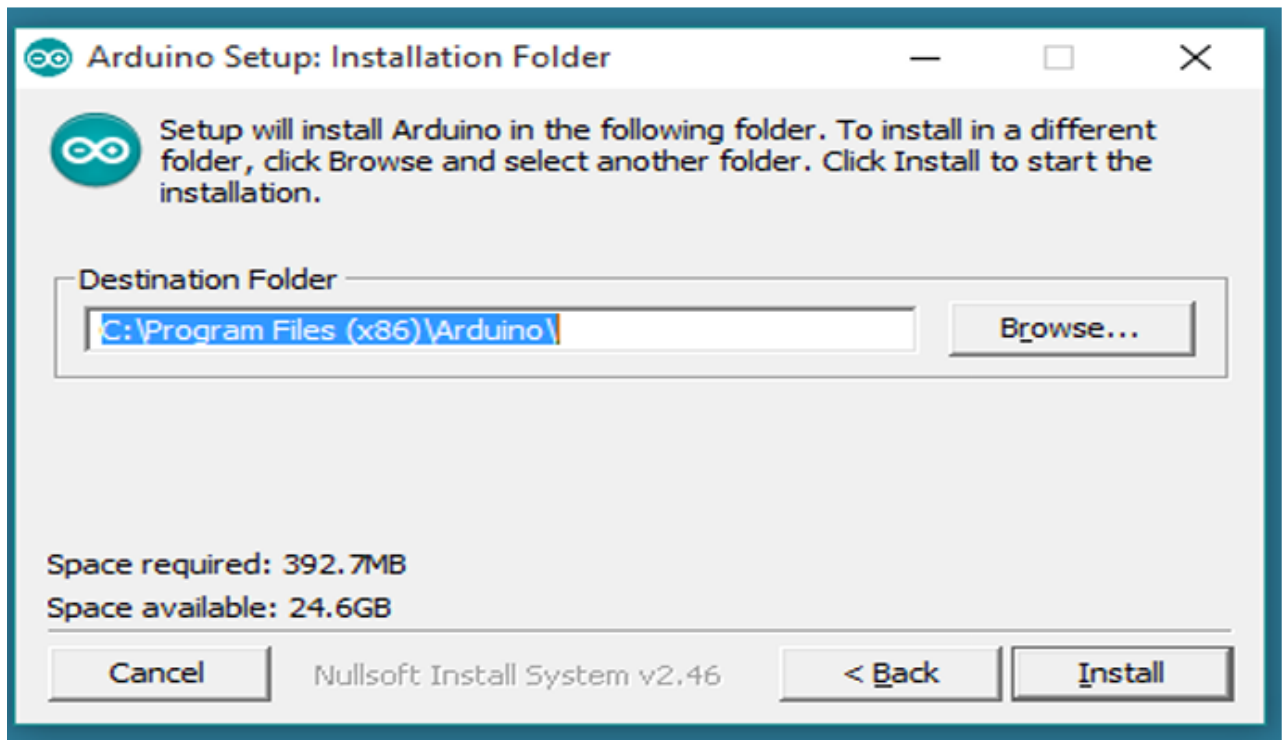
- Writing the program code for the ESP32 microcontroller.
- Compiling the code to detect syntax errors or logical issues.
- Uploading the compiled code to the ESP32 microcontroller.
- Monitoring the system output and debugging using the serial monitor.

Arduino IDE simplifies the development process and enables easy integration of hardware components used in the RFID-based inventory tracking system.

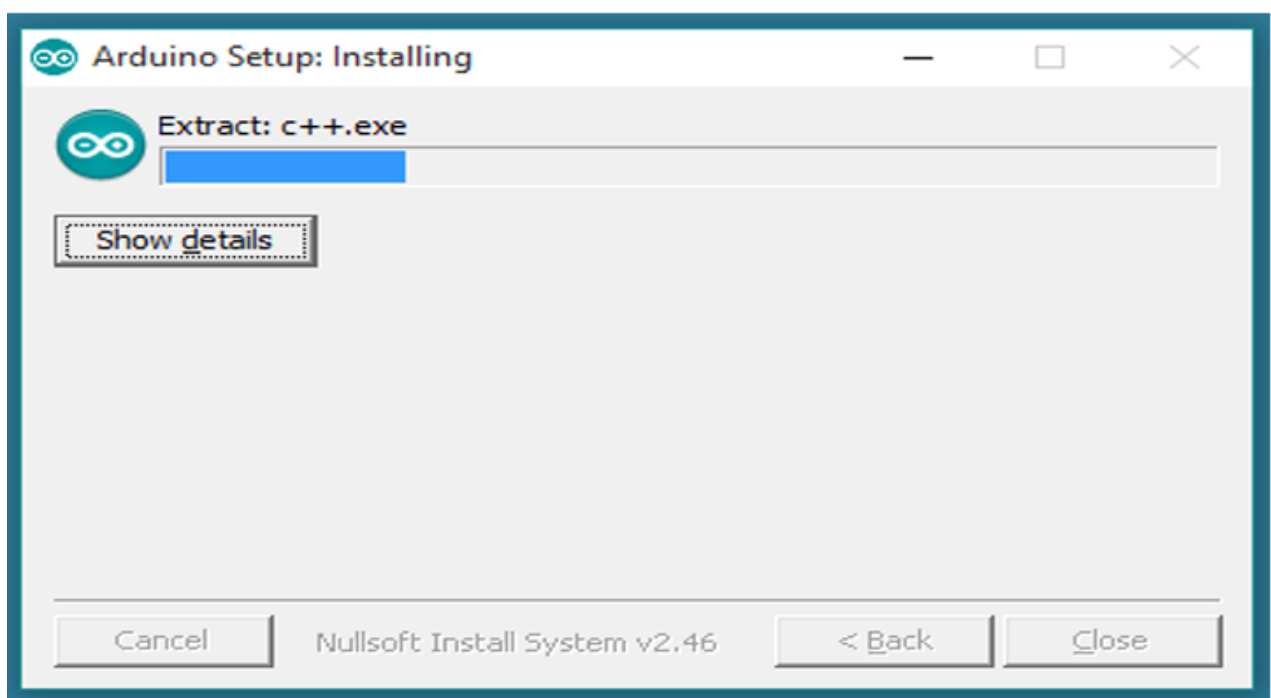
Get the latest version from the download page. You can choose between the Installer (.exe) and the Zip packages. We suggest you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually. The Zip file is also useful if you want to create a portable installation. When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.



Choose the components to install



Choose the installation directory (we suggest to keep the default one)



The process will extract and install all the required files to execute properly the Arduino Software (IDE)

6.2 EMBEDDED C

Embedded C is a programming language widely used for developing software for embedded systems. It is an extension of the C programming language specifically designed for programming microcontrollers and hardware-based systems.

In this project, Embedded C is used to program the ESP32 microcontroller, which acts as the central controller of the system. The program written in Embedded C controls the interaction between the RFID reader, LCD display, Wi-Fi module, and the cloud server.

The microcontroller reads data from the RFID reader whenever a medicine package with an RFID tag is scanned. The program then processes the tag information and sends the data to the cloud server through Wi-Fi connectivity. The system can also display system status messages on the LCD screen.

The main functions performed by the Embedded C program in this project include:

- Reading data from the RFID reader.
- Processing RFID tag information.
- Displaying system messages on the LCD display.
- Sending the scanned data to the cloud server through Wi-Fi.

Using Embedded C ensures efficient communication between the hardware components and enables reliable operation of the entire system.

SOURCE CODE

```

#define BLYNK_TEMPLATE_ID          "TMPL3Swyg6gFh"

#define BLYNK_TEMPLATE_NAME        "Quickstart Device"

#define BLYNK_AUTH_TOKEN          "wddajQ0ZXX0HjYfEpryPbq-Of2UyAQz2"

#define BLYNK_PRINT Serial

#include <WiFi.h>

#include <WiFiClient.h>

#include <BlynkSimpleEsp32.h>
char ssid[] = "USER_NAME";

char pass[] = "PASSWORD";
BlynkTimer timer;

BLYNK_WRITE(V0)

{

  int value = param.asInt();
  Blynk.virtualWrite(V1, value);

}

BLYNK_CONNECTED()

{

  Blynk.setProperty(V3, "offImageUrl", "https://static-image.nyc3.cdn.digitaloceanspaces.com/general/fte/congratulations.png");

  Blynk.setProperty(V3, "onImageUrl", "https://static-image.nyc3.cdn.digitaloceanspaces.com/general/fte/congratulations_pressed.png");

  Blynk.setProperty(V3, "url", "https://docs.blynk.io/en/getting-started/what-do-i-need-to-blynk/how-quickstart-device-was-made");

```

```

}
void myTimerEvent()
{
    Blynk.virtualWrite(V2, millis() / 1000);
}
void setup()
{
    Serial.begin(115200);
    Blynk.begin(BLYNK_AUTH_TOKEN,  ssid,  pass);
    timer.setInterval(1000L, myTimerEvent);
}
void loop()
{
    Blynk.run();
    timer.run();
}

```

6.2.1 FEATURES OF EMBEDDED C

Embedded C offers several advantages for developing embedded systems and hardware-based applications.

Some of the important features of Embedded C are listed below:

- **Efficient Hardware Control**

Embedded C provides direct access to hardware components such as ports, registers, and peripherals, allowing precise control of microcontroller operations.

- **Fast Execution Speed**

Programs written in Embedded C execute quickly, which is important for real-time applications such as RFID-based inventory tracking systems.

- **Low Memory Usage**

Embedded C programs are optimized for systems with limited memory and processing power, making them suitable for microcontroller-based applications.

- **Easy Integration with Peripherals**

Embedded C allows easy interfacing with hardware devices such as sensors, displays, communication modules, and RFID readers.

These features make Embedded C a suitable programming language for implementing the software logic of the proposed system.

6.3 CLOUD DATABASE AND WEB DASHBOARD

The cloud platform plays an important role in storing and managing the inventory data generated by the RFID system. When the ESP32 microcontroller reads RFID tag information, the data is transmitted to the cloud server through Wi-Fi connectivity.

The cloud database stores the received data and maintains records of scanned RFID tags, medicine details, and inventory updates. The database automatically updates the stock levels whenever new RFID data is received.

A web-based dashboard is developed to display the stored inventory information in an organized and user-friendly format. The dashboard allows hospital administrators and pharmacy staff to monitor the medicine inventory remotely.

The web dashboard displays the following information:

- Current medicine stock levels.
- Recently scanned medicines.
- Real-time inventory updates.

This system helps administrators easily monitor medicine availability and take necessary actions when stock levels are low or medicines are close to expiry. The use of a cloud database and web dashboard improves transparency, reduces manual effort, and ensures efficient inventory management in hospital pharmacies.

CHAPTER 7

RESULT

The developed RFID-based inventory tracking system successfully demonstrates automated monitoring of medicines in hospital pharmacies. RFID tags attached to medicines are detected automatically by RFID readers, and the data is transmitted to the cloud platform through the ESP32 microcontroller.

The system displays real-time inventory information on the web dashboard and LCD display. The results show improved accuracy in tracking medicines and reduced manual effort in managing pharmacy inventories.

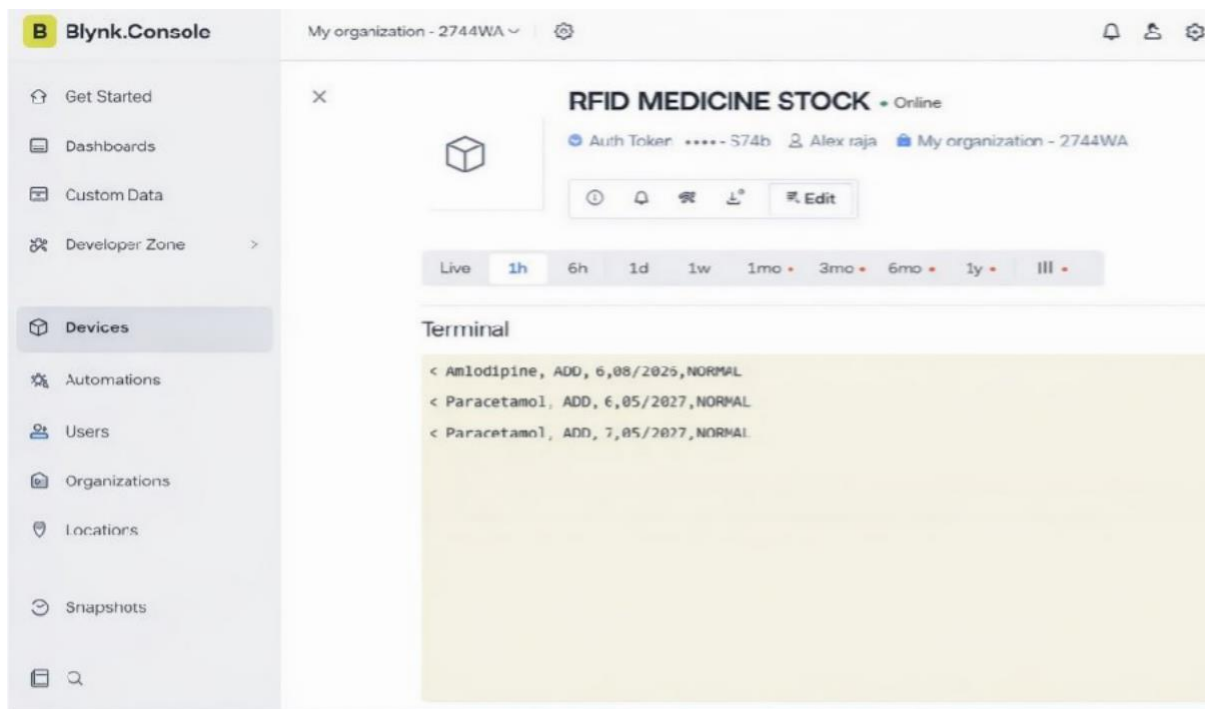


Fig 7.1 Output 1

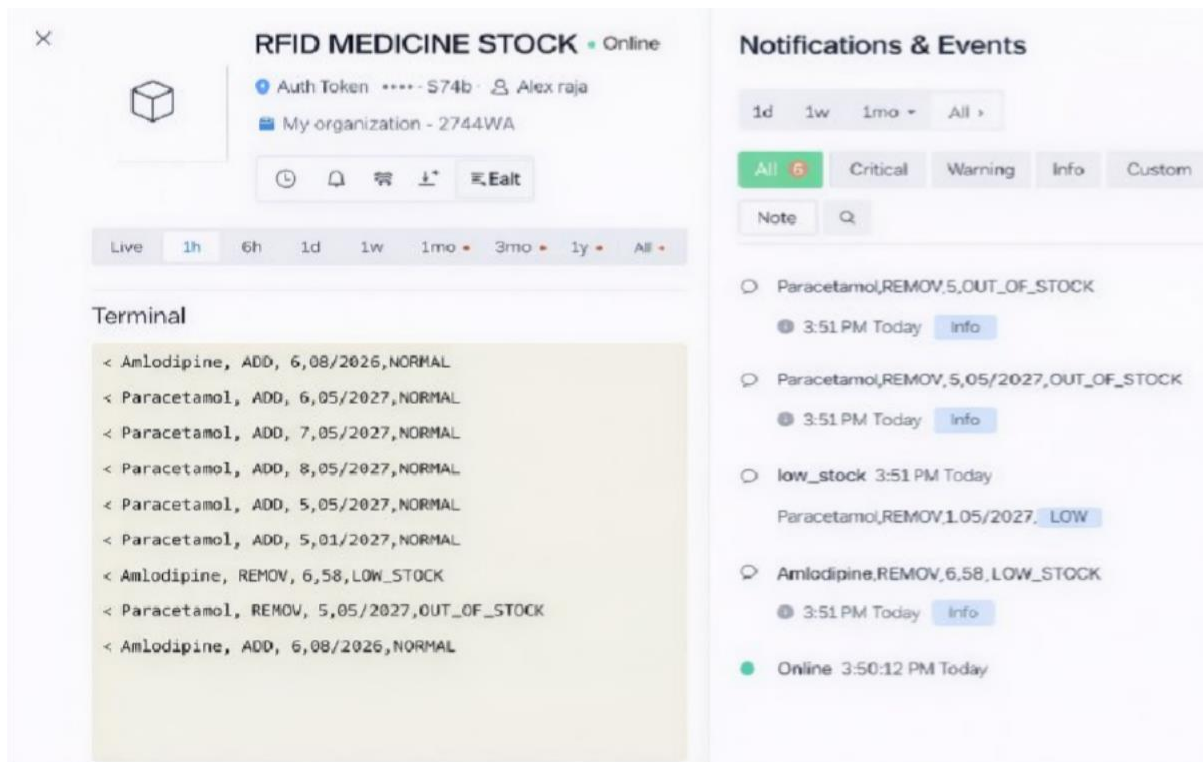


Fig 7.2 Output 2

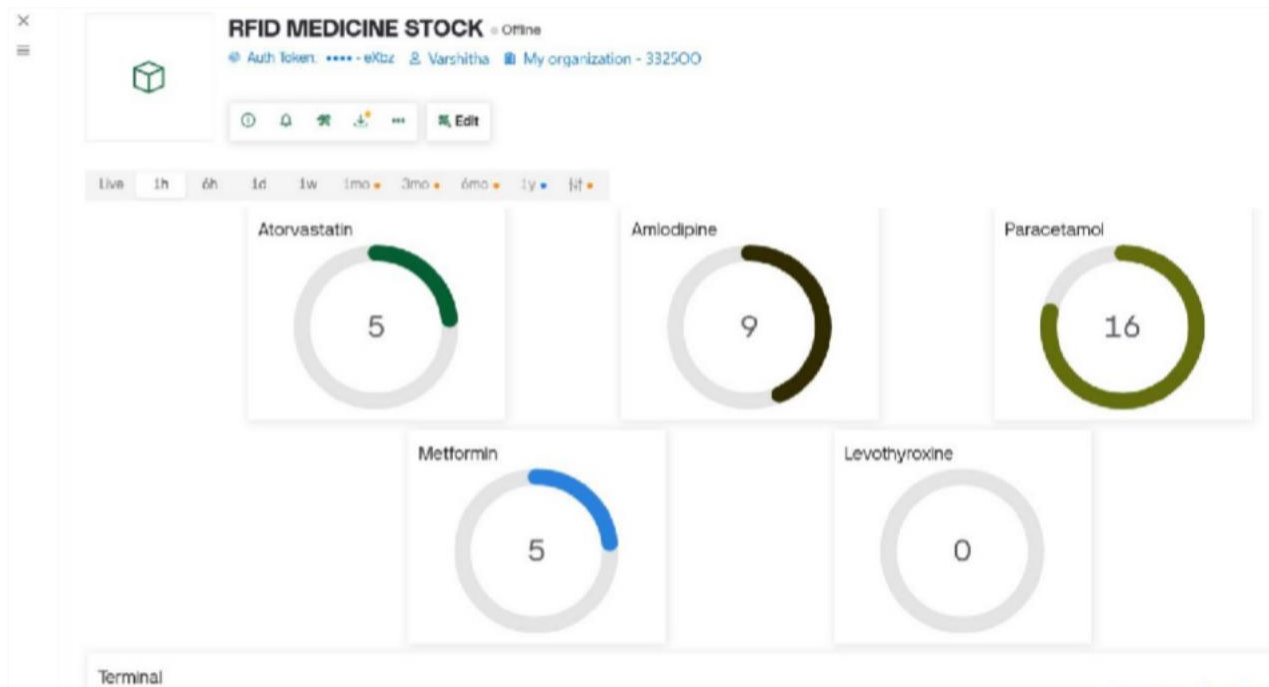


Fig 7.3 Output 3

CHAPTER 8

CONCLUSION

The project titled “**An Intelligent RFID and Cloud-Based Inventory Tracking System for Healthcare Supply Chains**” successfully demonstrates the implementation of an automated and efficient solution for managing medicine inventory in hospital pharmacies.

In traditional healthcare inventory systems, manual record-keeping and barcode-based tracking methods often lead to inaccuracies, delays, and inefficiencies. These issues can result in serious consequences such as medicine shortages, overstocking, and wastage due to expired drugs. The proposed system addresses these challenges by integrating **RFID technology, IoT communication, and cloud-based data management** to provide a real-time, automated inventory monitoring solution.

In this system, RFID tags attached to medicine packages enable automatic identification and tracking without the need for manual scanning. RFID readers detect the movement of medicines, and the ESP32 microcontroller processes the data and transmits it to a cloud server using Wi-Fi connectivity. The cloud platform stores and updates the inventory data, which can be accessed through a web-based dashboard.

The developed system achieves the following key outcomes:

- **Real-time Inventory Tracking:** The system continuously monitors medicine movement and updates stock information instantly.
- **Reduction of Manual Errors:** Automated RFID scanning eliminates human errors associated with manual data entry.
- **Improved Efficiency:** The system reduces the time required for inventory

management and auditing.

- **Remote Accessibility:** Cloud integration allows administrators to monitor inventory from any location.
- **Better Stock Management:** The system helps in maintaining optimal stock levels and avoids shortages.

The implementation of this system proves that RFID and IoT technologies can significantly improve healthcare supply chain management. The system ensures better transparency, accuracy, and efficiency in tracking medicines, which ultimately contributes to improved patient care and hospital operations.

Although the current system focuses on real-time tracking and monitoring, it provides a strong foundation for further enhancements and integration with advanced technologies. Overall, the project successfully meets its objectives and demonstrates a practical application of modern technologies in healthcare inventory management.

CHAPTER 9

FUTURE ENHANCEMENT

The proposed RFID and cloud-based inventory tracking system can be further enhanced by incorporating additional features and advanced technologies to improve its functionality, scalability, and usability.

Some of the possible future enhancements are:

1. Mobile Application Integration

A mobile application can be developed to provide real-time access to inventory data. Hospital staff and administrators can receive notifications and monitor stock levels directly from their smartphones. This will improve accessibility and allow quicker decision-making.

2. Integration with Hospital Management Systems

The system can be integrated with existing hospital management software to create a unified platform. This will enable seamless coordination between different departments such as pharmacy, billing, and patient management systems.

3. Automated Billing and Checkout System

The system can be extended to include an automated billing module. When a medicine is dispensed and scanned using RFID, the billing system can automatically update patient records and generate invoices, reducing manual effort.

4. Expiry Date Alert System

An advanced feature can be added to track the expiry dates of medicines. The

system can generate alerts when medicines are nearing expiry, helping to reduce wastage and improve inventory utilization.

5. Multi-Location Inventory Management

The system can be expanded to manage inventory across multiple hospital branches or medical stores. A centralized cloud platform can provide a unified view of inventory across different locations.

6. Enhanced Security Features

Security can be improved by implementing authentication mechanisms such as user login systems, role-based access control, and encrypted data transmission to protect sensitive inventory data.

7. Integration with Barcode Systems

For compatibility with existing systems, barcode scanning can be integrated alongside RFID technology. This hybrid approach ensures flexibility and ease of adoption in different healthcare environments.

8. Data Analytics and Reporting

The system can include advanced reporting features to analyze inventory trends, usage patterns, and stock movement. These reports can help administrators make better decisions regarding procurement and stock management.

9. Notification and Alert System

Automatic alerts can be sent via SMS or email for low stock levels, stock updates, or system errors. This ensures timely action and improves operational efficiency.

10. Scalable Cloud Infrastructure

The cloud system can be enhanced to support large-scale deployment with improved performance and reliability. This allows the system to handle data from multiple hospitals and large inventories efficiently.

CHAPTER 10

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